Engineering Applications of Artificial Intelligence 151 (2025) 110632

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Aspect-based sentiment analysis for software requirements elicitation using fine-tuned Bidirectional Encoder Representations from Transformers and Explainable Artificial Intelligence

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ARTICLE INFO

Keywords:
Software requirement elicitation
Sentiment analysis
Aspect Category Detection
Aspect Category Polarity
App reviews
Fine-tuned Bidirectional Encoder Representations from Transformers
Explainable Artificial Intelligence

ABSTRACT

Aspect-Based Sentiment Analysis (ABSA) of app reviews allows a better understanding of user preferences regarding specific product features and helps the development team elicit requirements effectively. The existing literature faces challenges such as limited focus on the automation of Requirement Elicitation (RE), insufficient task-specific fine-tuning of models such as Bidirectional Encoder Representations from Transformers (BERT), and lack of interpretability owing to the black-box nature of these models. Therefore, our work makes the following significant contributions to address these challenges: (1) development and evaluation of a robust method based on ABSA for the automation of the RE process; (2) optimization of ABSA using BERT fine-tuning for enhanced performance, which includes conducting a comprehensive ablation study to obtain the best hyperparameters that guarantee the best model performance and robustness; and (3) integration of Explainable Artificial Intelligence (XAI) techniques for enhanced BERT model interpretability. Our work was evaluated on the ABSA Warehouse of Apps REviews (AWARE) dataset, a specifically tailored dataset for the RE process. Our study outperformed baseline models such as the Support Vector Machine (SVM), Convolutional Neural Network (CNN), and BERT, and achieved an average F1-Score of 0.83 for the Aspect Category Detection (ACD) task and 0.94 for the Aspect Category Polarity (ACP) task. In addition, we employed XAI using Locally Interpretable Model-Agnostic Explanations (LIME) to explain the BERT model prediction results, which aids in the improved visualization and interpretability of the app review analysis for the automated RE process.

1. Introduction

App reviews are user-generated feedback in the form of text that is usually available on App Store[1] and Google Play[2] for different mobile apps. App reviews allow users to express their opinions and preferences regarding mobile apps. App reviews generally provide feedback about existing app features or aspects, such as performance, ease of use, usability, bugs, or requests for new features.

The landscape of mobile device application development is rapidly evolving. User feedback in the form of app reviews is the most important factor behind the success of any mobile application. These app reviews serve as a direct channel through which the software development team can better comprehend user satisfaction (Al-Subaihin et al., 2019). The development team can also identify areas for improvement in mobile applications by analyzing app reviews. App reviews provide a development team with a feedback loop that promotes continuous improvement and innovation in mobile application development (Dkabrowski et al., 2022). App reviews serve as a guide for the development team to prioritize features for the next update (Dkabrowski, 2022). App reviews are a valuable data source that may be used to develop user-centered experiences that put the needs of the user upfront with the development team and stakeholders.

To gain insights from app reviews for Requirement Elicitation (RE), the development team must analyze the review text. App reviews contain a large amount of text, and thousands of reviews are generated daily by users online. The traditional approach for app review analysis is manual, in which human evaluators carefully analyze and categorize app reviews. This manual analysis of app reviews can be difficult for the development team because it is a time-consuming, labour-intensive

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[1] https://apps.apple.com/us/genre/ios/id36
[2] https://play.google.com/store/games?hl=en

https://doi.org/10.1016/j.engappai.2025.110632
Received 10 July 2024; Received in revised form 23 February 2025; Accepted 17 March 2025
Available online 27 March 2025

task, and has the potential to introduce bias while evaluating user feedback (Nakayama, 2013; Park et al., 2018).

Recent advances in Machine Learning (ML), Deep Learning (DL), and Natural Language Processing (NLP) have introduced automated analysis of app reviews using Sentiment Analysis (SA). SA is a specific technique dedicated to the extraction of sentiments from a given piece of text, which is usually a review or blog post for a product or service (Lamba and Madhusudhan, 2021).

SA can be performed using two main approaches: lexicon-based and machine-learning-based (Taj et al., 2019b). The Lexicon-based approach uses a lexicon dictionary containing vocabulary words and their associated polarity scores. This approach classifies given reviews as positive, negative, or neutral by analyzing the polarity of words in the review text. Lexicon-based approaches are simple and efficient to implement; however, there are issues with this approach, such as limited vocabulary, out-of-vocabulary words, and lack of contextual understanding. In a machine-learning-based approach, a model is trained to learn linguistic patterns and semantic features from reviews to predict the sentiments hidden in the reviews.

SA can be performed at various levels of abstraction, such as document level, sentence level, phrase level, and aspect level (Wankhade et al., 2022). Document-level SA aims to analyze the entire text in a document to determine its polarity. Documents can be social media blogs, articles, or reviews. Sentence-level SA focuses on determining the polarity of individual sentences in a document. Even if we want to analyze it in a more granular manner, we can use phrase-level SA. This level of SA aims to identify sentiment-bearing phrases and expressions in individual sentences. Overall, the SA task lacks granularity when analyzing the review text.

Aspect-level Sentiment Analysis, also known as Aspect-Based Sentiment Analysis (ABSA), is an NLP technique that is more fine-grained. ABSA is designed to analyze text data and extract opinions or sentiments towards specific features or aspects of a product or service (Schouten and Frasincar, 2015). Generally, ABSA tasks are further divided into two subtasks: Aspect Category Detection (ACD) and Aspect Category Polarity (ACP) (Pontiki et al., 2014).

1. **ACD Task:** The main goal of this task is to classify text or reviews in a pre-defined set of categories or categories extracted from review text. Categories refer to the specific characteristics of the entity being discussed or evaluated. For example, if we analyze the app review “I love the audio button; it allows me to have conversations with my friends, and it loads as fast as regular messages.” obtained from Aspect-Based Sentiment Analysis on the ABSA Warehouse of Apps REviews dataset (AWARE) dataset. The aspect category for this review is “usability”. The aspect categories given in the AWARE dataset are predefined, designed, and annotated from a requirement elicitation point of view. An ACD task can be performed in a supervised or unsupervised manner (Zhang et al., 2022). In supervised ACD, annotated labels are required to train the model. However, in unsupervised ACD, there is no need for annotated labels to train the model.
2. **ACP Task:** The main goal of this task is to assign sentiment polarity to each review’s aspect category. The sentiment value can be positive, negative, or neutral based on the sentiment expressed in that review. For example, if we have an app review from the AWARE dataset, “There seem to be multiple bugs after the most recent update.” First, the model performs an ACD task and then assigns a polarity value for this review. In this case, it is assigned a “Negative” polarity.

ABSA has attracted considerable attention, especially for analyzing user feedback information, owing to its advantages in extracting fine-grained sentiment features. The existing literature on ABSA based on conventional NLP or ML methods and those based on DL models, such as BERT, has shown that ABSA approaches can improve the performance of sentiment analysis, aspect term extraction, and aspect category detection (Do et al., 2019; Chauhan et al., 2023). However, these models are generally black-box in nature making it hard to understand why a particular decision has been made. This lack of interpretability poses a serious problem for RE because it is crucial to comprehend the underlying reasoning behind these predictions in addition to identifying specific features or perspectives. To fill this gap, we present a new approach that integrates a BERT-based ABSA model and Explainable Artificial Intelligence (XAI). The Locally Interpretable Model-Agnostic Explanations (LIME) (Ribeiro et al., 2016) XAI method is introduced in the proposed model for instance-level explanations. LIME explains words or phrases that primarily influence model decisions regarding sentiment and aspect-category predictions. Our proposed approach not only makes the ABSA tasks more robust and transparent but also provides developers with effective information for finding new requirements for correction or improvement of software apps from user comments or app reviews.

1.1. Research gap

This section identifies three major research gaps that lay the foundation for our proposed work.

1. **Limited Focus on RE Automation:** The existing literature suggests that despite the various potential benefits of using NLP-based techniques to automate the RE process from app review analysis, there is a lack of research utilizing NLP-based techniques for app review analysis to automate the RE process based on specific aspects.
2. **Need for Task-Specific BERT fine-tuning:** Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2018) is pre-trained using unsupervised learning on a very large corpus for basic tasks such as masked language modeling and next-sentence prediction. To use BERT for various downstream tasks, such as Named Entity Recognition (NER), question answering, or sentiment analysis, it is necessary to fine-tune the BERT with training data for the desired task domain. Task-specific fine-tuning of the BERT model remains underexplored in certain applications such as RE automation.
3. **Lack of Interpretability in BERT Models Decision-Making:** DL models such as BERT, RoBERTa (Robustly optimized BERT approach) (Liu et al., 2019) and GPT (Brown et al., 2020) have achieved impressive performance results on various NLP tasks. However, owing to the black-box structure, there is always an element of risk when deploying DL models. A major risk found in the literature for these DL models is bias, which can be depicted in several ways. It is obvious that this bias in DL models is a statistical bias resulting from the bias in the real world and historical data on which the model is trained (Parikh et al., 2019). Because of this behavior, it is not suitable for humans to over-rely on the automated decision-making of DL models.

1.2. Research contribution

Our work is motivated by the necessity to use app reviews, a useful source of user feedback for software development teams, to automate the RE process. The existing literature suffers from limitations, such as a limited focus on RE automation, inadequate task-specific fine-tuning of models such as BERT, which results in non-robust performance, and the black-box nature of deep learning models such as BERT, which makes predictions difficult to interpret. To address these limitations, we proposed a unique approach to streamline the automated RE process and obtain deep insights into user preferences and requirements by improving the performance and robustness of the BERT model for ABSA tasks. Our work also introduces LIME explanations for better interpretation of app review analysis. Our study makes three significant contributions to the field of NLP, specifically in the domain of ABSA, by automatically analyzing app reviews for the RE process.

1. **Development and evaluation of a robust method based on ABSA for automation of RE process:** The first contribution of our work is the development and evaluation of an ABSA-based method for app review analysis to automate the RE process, as the existing literature has limited focus on leveraging NLP-based techniques for the RE task. Our proposed method outperforms state-of-the-art (SOTA) methods on the AWARE dataset.
2. **Optimization of ABSA using BERT fine-tuning for enhanced performance:** The second contribution focuses on fine-tuning of the BERT model for optimized ABSA tasks, in which we conducted a comprehensive ablation study to determine the best hyperparameters for BERT fine-tuning. This fine-tuning not only helps the BERT model in task-specific adaption but also improves its performance by leveraging its learning from both the pre-training and fine-tuning processes.
3. **Integration of XAI techniques for enhanced BERT interpretability:** The third contribution entails improving the decision-making process of the BERT model by integrating the XAI. Based on our experimental findings, the integration of XAI improves the explainability of the BERT prediction for the ABSA tasks, making the model outcomes more comprehensible.

The remainder of this paper is organized as follows. Section 2 provides a literature review on BERT-based ABSA, XAI, and NLP for requirement elicitation. Section 3 presents the proposed methodology and Section 4 discusses the results. Section 5 highlights the limitations of this study and future work. Section 6 presents the conclusions of the study.

2. Literature review

Businesses and organizations have recently become interested in NLP because of their ability to represent and analyze human language computationally. Its applications have expanded to include machine translation, email spam detection, app review analysis, ABSA, information extraction, text summarization, and synthetic text generation (Khurana et al., 2023). Businesses are able to derive actionable insights from customer reviews, comments, and social media postings by using ABSA, one of the most effective NLP techniques for extracting comprehensive opinions from textual data. This section provides a literature review of related works related to ABSA, using NLP in the RE process, and the use of XAI to enhance the transparency and expandability of NLP, ML, and DL models. The integration of these methods is also examined, emphasizing how they can be used in conjunction to overcome the difficulties in deriving interpretable and useful insights from user input for the RE process. Early research on ABSA has relied on unsupervised, semi-supervised, and rule-based approaches. Initially, techniques such as clustering, information extraction, data mining, and summarization were used for ABSA tasks, in which researchers were interested in finding opinion words and the polarity of review sentences. Hu and Liu (2004) introduced an opinion-based summarization system for customer review analysis in which the system first mines the product features from customer reviews using data mining, then Part-of-Speech (POS) tagging is used to find opinion words from review sentences, and WordNet is used to find the semantic orientation of review sentences. Ultimately, the system generates a summary of opinion words for specific product features and polarities. In ABSA, one of the common subtasks is to find aspect terms. Unsupervised approaches have also been popular in the past because of their ease of use and flexibility. Therefore, labeled data are not required for Aspect Term Extraction (ATE) and polarity detection. Moghaddam and Ester (2010) have introduced an unsupervised approach for unstructured reviews to find aspect terms and aspect ratings. Many studies in the literature have used the POS tagging technique to extract aspect terms from review texts. Topic modeling techniques such as Latent Dirichlet Allocation (LDA) have also been widely used for ABSA tasks, specifically for ACD tasks (Poria et al., 2016). Fu et al. (2013) used word-level analysis and proposed a data-mining technique to highlight the impact of individual words on user sentiments expressed in app reviews. The authors also used LDA as a text analysis technique for topic modeling to identify the reasons for user dissatisfaction with particular app features. Semi-supervised approaches to ABSA have also been explored. Haq et al. (2023) proposed a novel semi-supervised approach for ATE and Aspect Category Detection (ACD). The authors used topic modeling to generate aspect categories along with contextual augmentation to enhance aspect categorization, and Zero-Shot Learning (ZSL) to classify aspect categories.

Traditional ABSA approaches have contributed significantly to the field, despite being fundamental, these approaches have a number of limitations that reduce their usefulness in complex, practical applications. Traditional approaches to ABSA mostly depend on predefined rules, lexicons, and machine learning algorithms, such as Support Vector Machines (SVM) and Naïve Bayes, for which manual feature engineering is required. Because of this dependence, these approaches require modification or adjustment for additional datasets, making them less adaptable and more challenging to generalize across domains. In addition, these techniques have trouble handling implicit aspects, which are sentiments stated without specifically referring to the target aspect. Their inability to comprehend sophisticated linguistic structures, such as irony, sarcasm, and negation, also causes them to make mistakes when classifying sentiments and aspects. Furthermore, traditional ABSA approaches fail to represent contextual dependencies and semantic linkages between words, which are essential for accurately interpreting sentiments and aspects. To overcome these limitations, advanced approaches are required, such as deep learning and transformer-based models, which can use contextual embeddings and attention mechanisms to classify sentiments and aspects robustly (Nazir et al., 2020).

Irsoy and Cardie (2014) used for the first time deep learning for Opinion Term Extraction (OTE) using deep Recurrent Neural Network (RNN). The study treated this task as token-level sequence labeling. RNN outperformed existing approaches for OTE, such as semi-Markov conditional random fields (semi-CRFs). Advanced approaches to ABSA often use pre-trained word embedding, such as Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014). However, these pre-trained word embeddings suffer some challenges as they are context-independent; therefore, due to this reason, understanding complex dependencies is difficult. Additionally, these word embeddings cannot handle out-of-vocabulary (OOV) words owing to their fixed vocabulary. In addition to the existing machine learning and deep learning approaches, the introduction of Large Language Models (LLMs), such as BERT (Devlin et al., 2018) and RoBERTa (Liu et al., 2019) has brought substantial improvements to NLP tasks, particularly ABSA, in recent years. The word embedding produced using these LLMs is highly effective in capturing long-range contextual dependencies in text and is effective for various NLP tasks. Li et al. (2019) introduced a novel approach for end-to-end ABSA in BERT fine-tuning. Specifically, the authors used a linear classification layer on top of the BERT model with a unified tagging scheme, which helped achieve promising results. According to this approach, the classification model uses the review sentence as input and can identify both aspects and aspect sentiments from a given review text. The efficiency of BERT in enhancing aspect extraction precision while preserving scalability across many domains is highlighted by Akram and Sabir (2023). The Authors also emphasized on the use of pre-trained transformer-based models such as BERT and their proper fine-tuning for handling contextual dependencies which make them suitable for the modern ABSA tasks. The fine-tuning flexibility of BERT helps in domain-specific adaptation, which enhances its robustness and efficiency. These studies highlighted the robustness of the BERT model for ABSA tasks. Hoang et al. (2019) also highlighted that the BERT model works very well for ABSA tasks because of its capacity to manage polysemy, model long-range dependencies, and collect contextual information. The advantages mentioned above enabled us to use the BERT model and introduce improvements for the ABSA task through fine-tuning.

Traditional NLP, ML, and DL models are black-box and lack inherent transparency. Owing to a lack of transparency, the decision-making processes of these models are difficult to interpret. This results in a lack of trust, accountability, and usability issues.

To address the problem of the interpretability of these black-box models, a new set of methods called Explainable AI (XAI) has been developed. Integrating XAI into traditional NLP, ML, and DL models bridge the gap between the black-box nature and demand for transparency, interpretability, and accountability in these models (Phillips et al., 2020). Mathews (2019) highlighted the application of XAI to NLP. The author conducted a case study on the importance of interpretability in NLP tasks. So (2020) emphasized the need to integrate SA and XAI to promote the interpretability of rating prediction models and determine how the emotion expressed in user reviews impacts the ratings. The author employed machine learning methods, namely, Random Forest, and XGBoost. For XAI, the author used (1) Feature Importance: to calculate how much a feature affects predictions across the globe, and (2) Local Attribution to illustrate every feature's instance-level engagement. (3) Partial Dependency Plot (PDP): this illustrates the importance of individual features within their range by estimating the changes in the target variable. This approach allows for the identification of biases, incorrect prediction processes, and ideas for developing robust models.

XAI methods such as LIME and SHAP (SHapley Additive exPlanations) (Lundberg, 2017) provide localized explanations in terms of the contribution of each feature of an input to the model's prediction, thereby increasing interpretability and comprehensibility of the behavior of a model. These methods have been successfully used in fields such as text classification and sentiment analysis to improve the interpretability of the models. Malhotra et al. (2021) performed experiments on tweet sentiment analysis and introduced XAI using the LIME technique to interpret the classification results visually. This study highlights how the incorporation of LIME into the text sentiment analysis of tweets shows that using deep learning models coupled with XAI methods can provide both high accuracy and interpretability in sentiment analysis. Zhao et al. (2020) proposed an approach named CNN-SHAP, for computing SHAP scores of the CNN-based text classification model. This approach builds on SHAP's ability to identify the importance of local features. Datta et al. (2021) explored the structure and working of Neural Machine Translation (NMT) using XAI techniques like Seq-2-Seq-Vis and LSTM-Vis. Both techniques served to reveal how NMT was performed using deep neural networks. However, no prior work has employed XAI techniques for ABSA tasks for app reviews; thus, a significant research gap exists in integrating fine-grained sentiment analysis with explainability.

Literature related to NLP for RE focuses on using NLP techniques to automate various aspects of the RE process. Dalpiaz et al. (2018) proposed the starting concept of NLP4RE (Natural Language Processing for Requirements Engineering), in which the authors depict how NLP can be applied to several RE activities. This study demonstrates that NLP applications can assist with the automation and improvement of tasks such as requirement elicitation, analysis, prioritization, and validation, particularly when there is an overwhelming use of natural language requirements. McZara et al. (2015) identified and prioritized important requirements from large sets by analyzing and extracting dependencies from natural language requirements using constraint solvers and NLP tools. To formalize natural language requirements, Sudhi et al. (2023) suggested an NLP-based pipeline. This study employed BERT-based Semantic Role Labeling (SRL) to identify semantic roles, such as agents and actions, that reduce manual rule creation for requirement formalization. Dependency parsing and POS tagging were used to analyze the sentence structure and extract logical relationships. This study successfully managed ambiguity and permitted scalable and adaptable requirements formalization, as demonstrated by tests conducted on industrial use cases. Dalpiaz and Parente (2019) proposed RE-SWOT an NLP-based tool to extract features and sentiment from app reviews using POS tagging, collocation patterns, and semantic similarity. By comparing the reference app with a competitor's app, features were categorized into SWOT categories (Strengths, Weaknesses, Opportunities, and Threats), facilitating needs elicitation through an interactive visualization tool. To compare competitive products, Jin et al. (2016) extracted typical and contrasting sentences from product reviews. The authors categorized similar subjects through clustering, determined the features of products via POS tagging, and determined the sentiment of opinions through sentiment analysis. The approach was validated on Amazon reviews to show that it works when conducting a competitor analysis. Baseline studies such as those by Alturaief et al. (2021) introduced the AWARE dataset of app reviews, specifically tailored from the RE perspective. The authors contributed to creating a benchmark for the ABSA-based RE process. They also set the baselines for the ABSA on the AWARE dataset using an SVM. Alturayeif et al. extended their work in Alturayeif et al. (2023) and achieved comparatively good results using the BERT model. Another attempt at an ABSA-based RE process was made by Gunathilaka and De Silva (2022). The authors employed CNNs to classify aspect categories as well as sentiments in reviews of mobile apps. They used two CNN models for aspect category classification and another for sentiment classification, both using pre-trained embeddings similar to Word2Vec, GloVe, and FastText. BERT-based contextual augmentation and round-trip translation procedures were used to handle class imbalances. The models thereby noted a marked improvement in their classification accuracy by utilizing the textual features as extracted by convolutional layers, max-pooling, and dense layers with softmax activation. Although NLP has demonstrated potential for automating the RE process, its application has only been briefly investigated in a few studies.

Meske et al. (2022) highlighted the scope of XAI for fulfilling the task-related needs of different XAI stakeholders. In addition, researchers have identified that XAI has diverse objectives for various stakeholder groups. Developers consider XAI's potential for debugging and improving AI systems, while AI users benefit from XAI by validating the reasoning of AI systems with their reasoning.

The existing literature only focuses on performance improvement on ABSA tasks. Interpretability and explainability, which are essential for fostering trust in AI systems, have received little attention in the literature. Additionally, the integration of XAI methods with ABSA tasks, particularly in the context of RE process automation from user feedback or app reviews, has not been adequately investigated. In summary, we conclude in this section that the existing literature lacks focus on using NLP-based techniques to automate the RE process. The work on ABSA still requires performance improvement, and there is also an emerging need to integrate XAI for better interpretability of NLP and DL models.

3. Methodology

This section discusses the proposed methodology for ABSA of app reviews based on BERT fine-tuning and XAI. The abstract architecture of the proposed methodology is illustrated in Fig. 1. Initially, app reviews from the AWARE dataset are used as model inputs. The input data underwent several text preprocessing and tokenization steps to ensure compatibility with the BERT model for the ABSA task. Subsequently, XAI was integrated using LIME with the fine-tuned BERT model to enhance the interpretability of the BERT model predictions. The remainder of this section provides a comprehensive discussion of the necessary details and steps of the proposed methodology.

3.1. The AWARE dataset

The input to the proposed fine-tuned BERT model is the app-review dataset. We used the ABSA Warehouse of Apps Reviews (AWARE)

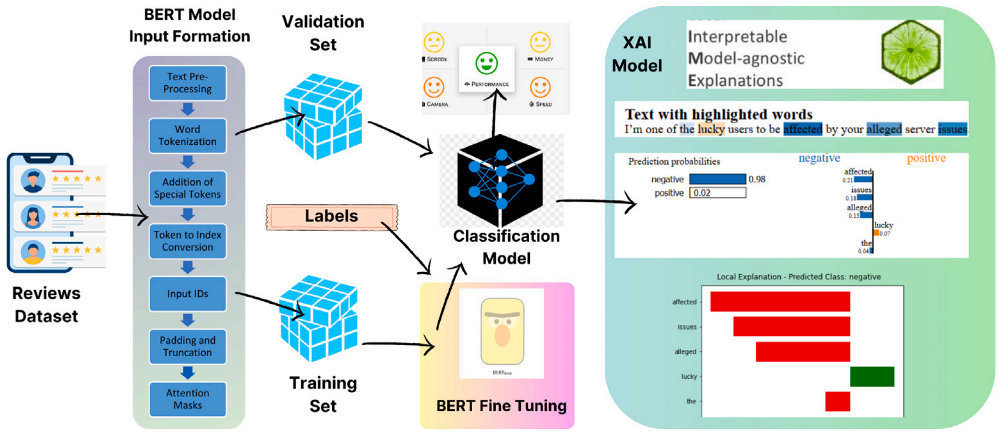


Fig. 1. Abstract architecture of proposed methodology.

dataset. The AWARE is a unique ABSA dataset of smartphone app reviews designed specifically to support the requirement-elicitation process.

Currently, the main focus of this research is on SOTA ABSA tasks on benchmark datasets (Chebolu et al., 2023) such as SemEval-2014 (Pontiki et al., 2014), SemEval-2015 (Pontiki et al., 2015), and SemEval-2016 (Pontiki et al., 2016). These datasets comprise reviews from laptops and restaurant domains. Sentihood contains question-answering data on urban neighborhoods (Saeidi et al., 2016), and the MAMS (Multi-Aspect Multi-Sentiment) dataset contains reviews from the restaurant domain (Jiang et al., 2019). However, one major drawback of these benchmark datasets is that the aspect categories do not represent the features of an application that could aid the development team in the requirement elicitation process. Owing to a lack of consideration for the needs of stakeholders when developing those datasets, particularly aspect categories in those datasets, they are not annotated or designed from a requirement elicitation perspective.

In addition to these datasets, a few other datasets are also tailored to support software development and requirement elicitation, such as the content categorization of tweet messages provided in Guzman et al. (2016) and the classification of requirements via the use of app reviews and user feedback in Guzman et al. (2015), Taj et al. (2019a), and Navarro-Almanza et al. (2017). However, none of these studies or datasets focused on aspect-based sentiment analysis tasks to automate the software evolution process. The AWARE dataset is the only ABSA benchmark dataset in which app reviews are specifically annotated from a requirement elicitation perspective, which is helpful for fine-grained analysis of app reviews. By using the AWARE dataset, researchers can perform Aspect Sentiment Classification or Aspect Category Polarity (ACP) task, Term Extraction, Aspect Category Classification or Aspect Category Detection (ACD), Sentiment Analysis, Explicit/Implicit Aspect Term Classification, Opinion/Not-Opinion Classification, and many other NLP tasks.

The AWARE dataset consists of 11,321 app review samples across three domains: (1) Social Networking, (2) Productivity, and (3) Games. It is divided into training, validation, and test sets to ensure proper model evaluation and generalization. For training and validation, 90% of the entire data is utilized (more specifically, 80% for training and 10% for validation). The 10% of the entire set of data was set aside as the test set in order to assess how well the model performed. Table 1 summarizes the training, validation, and test split for each domain of the AWARE dataset.

The sentiment counts for each domain, including Social Networking, Productivity, and Games, are shown in Fig. 2(a). Additionally, visualization of aspect categories concerning sentiment count is given in Figs. 2(b), 2(c), and 2(d) for Social Networking, Productivity, and Games domains. It is worth noting that the AWARE dataset has class imbalances for different aspect categories. The class imbalance can

Table 1
Train, validation, and test samples for each domain of the AWARE dataset.

Domain	Total	Train	Validation	Test
Games	4080	3264	408	408
Productivity	3846	3077	384	385
Social networking	3395	2716	339	340
Total	11321	9057	1131	1133

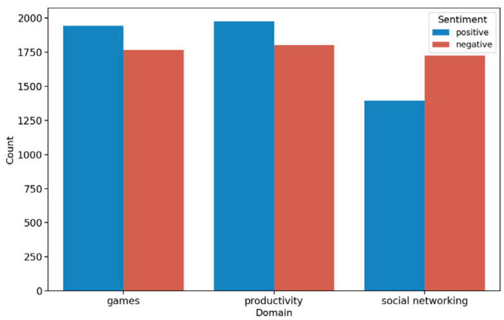
pose challenges during model training for ACD tasks as models can prioritize the majority class and ignore the minority class. In such cases, it is important to consider the class imbalance and ensure that the model performs well in all categories. To solve this, we tried the Synthetic Minority Over-sampling Technique (SMOTE) (Chawla et al., 2002) one of the popular methods for oversampling of the minority classes. Despite this, it can be observed that applying SMOTE did not lead to improved results of BERT. The BERT model is sufficiently robust for class imbalance problems (Kedas et al., 2022) owing to its better generalization and memorization avoidance capability (Tänzer et al., 2021).

3.2. BERT model

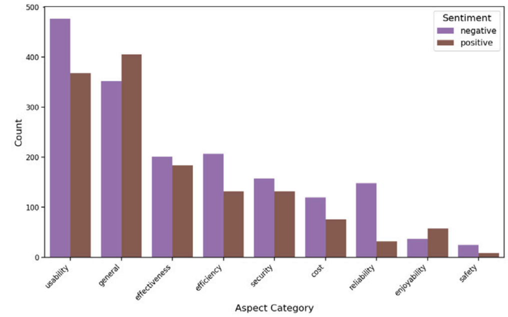
This section introduces our approach for ABSA subtasks, that is, ACD and ACP. We selected a BERT pre-trained model (Devlin et al., 2018). The BERT model was pre-trained on a large amount of unlabeled text data in a self-supervised manner during the training phase of the BERT. Because of the bidirectional nature of the BERT attention mechanism (Vaswani et al., 2017), it can capture complex word relationships and their dependencies better, as it considers the context of words in a sentence from both the left and right sides. BERT[3] has two variants, $\text{BERT}_{\text{base}}$ and $\text{BERT}_{\text{large}}$. For experimental purposes, we used the $\text{BERT}_{\text{base}}$ model because it is less parameter-intensive than $\text{BERT}_{\text{large}}$ and requires fewer memory resources than $\text{BERT}_{\text{large}}$. However, both variants of BERT are sufficiently robust to perform NLP tasks and provide almost identical performance results.

The $\text{BERT}_{\text{base}}$ architecture is based on 12 transformer blocks, the size of the hidden layers is 768, and the number of self-attention heads is 12. This pre-trained model was trained on 110M parameters (Wolf et al., 2019). The model framework consisted of two stages: *pre-training* and *fine-tuning*. In the pre-training stage, the model is trained on a very large, unlabeled corpus (Devlin et al., 2018), whereas in the fine-tuning stage, the model is initialized with pre-trained parameters and weights, and is trained using labeled data on a downstream task. A detailed diagram of the proposed BERT model is shown in Fig. 3.

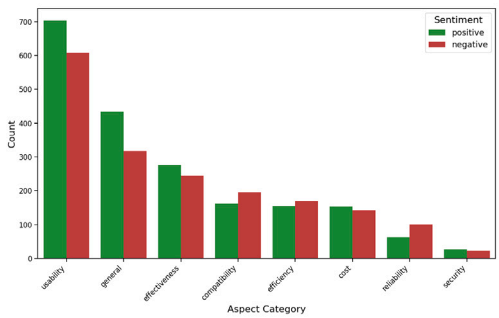
[3] https://github.com/google-research/bert



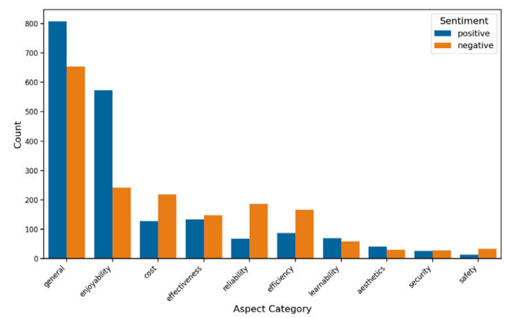
(a) Sentiment count for the AWARE dataset domains



(b) Aspect Category concerning sentiment count for the AWARE Social Networking Domain



(c) Aspect Category concerning sentiment count for the AWARE Productivity Domain



(d) Aspect Category concerning sentiment count for the AWARE Games Domain

Fig. 2. AWARE dataset exploratory data analysis.

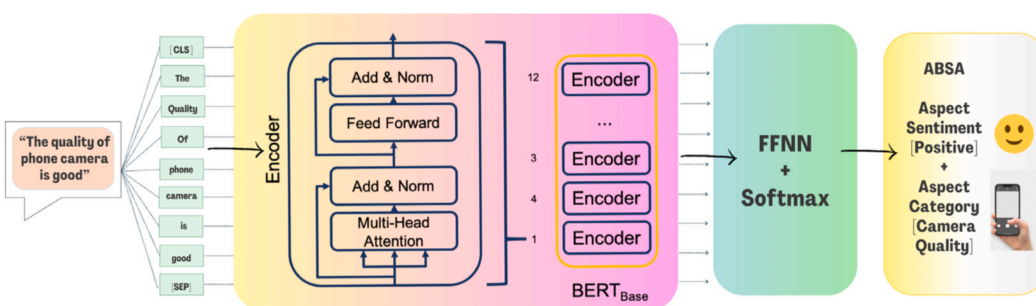


Fig. 3. Fine-tuned BERT architecture.

3.3. Model inputs

Input text data are prepared for the construction of the BERT model. Below, we summarize the input preparation process.

Text pre-processing: Text pre-processing refers to preparing text data before it can be used for any NLP task. The main goal of text pre-processing is to convert unstructured raw text into a structured form so that it can be easily understood and analyzed by any ML or DL model. Text preprocessing helps improve the efficiency of NLP tasks, such as information retrieval, text classification, and sentiment analysis performed by ML or DL models (Alam and Yao, 2019; Pradha et al., 2019; de Oliveira and Merschmann, 2021). Specifically, if we consider ABSA tasks, this process helps transform the raw text of app reviews into a structured form that facilitates the accurate classification of app review categories and sentiments. This process involves cleaning the text data by performing tasks such as removing punctuation, converting the text of app reviews to lowercase, removing stop words, and stemming or lemmatizing words. In our study, we also preprocessed the app review dataset to ensure that the input to the model was clean, structured, and normalized. The following steps are taken to perform text pre-processing on the AWARE dataset:

- **Case conversion:** The review text data is converted to lowercase to make it uniform and case-insensitive.
- **Removal of URLs:** URLs in review text are often irrelevant to NLP tasks. In this step, URLs were removed from the review text.
- **Removal of emojis:** Review text also contains emojis; for NLP tasks, it is necessary to remove them from review text. In this step, emojis are converted into textual representations.
- **Removal of special characters:** Special characters such as punctuation marks, symbols, and white spaces are non-numeric and non-alphabetic in nature. These special characters do not have any significant meaning in NLP tasks. Special characters were removed from the review text.
- **Tokenization of text:** Tokenization is a text pre-processing step in which text data is split into smaller chunks known as tokens. These tokens can be words, phrases, characters, or subwords. Tokenization helps in text classification and analysis tasks, where each individual token helps extract useful information from text data.
- **Lemmatization & Stemming:** In lemmatization and stemming, words are reduced to their root or base form. Lemmatization converts words into dictionary forms. In stemming, prefixes and suffixes are removed from words to convert them into base forms.

- **Word tokenization:** The BERT model considers the input text as a sequence of tokens. The first step in preparing the BERT model input is word tokenization. The BERT tokenizer is used to split pre-processed text into tokens. The WordPiece tokenization approach is used by the BERT tokenizer. According to this tokenization approach, each word from a given sentence is further divided into subwords or long pieces of 3–4 characters. This helps enhance the BERT model performance, as some rare words might not be present in the training set.
- **Addition of special tokens:** Afterword tokenization, BERT uses special tokens such that the model can easily understand the structure of the input sentence. The common use of special tokens is to mark the boundaries of the given sentences and unknown tokens. A [CLS] token is used to mark the beginning of sentences. It is also used to mark the classification head for a given input sentence, which helps predict the output label of a given text. The [SEP] token marks the end of the sentence or separates the different sentences from the given input sequences. Occasionally, unknown words or out-of-vocabulary words (OOV) occur in the input sequence of the model. To handle such cases, the [UNK] token is used in the tokenization to represent unknown words.
- **Token to index mapping:** After adding special tokens, the next important step is to map all tokens to their corresponding index values from the model's vocabulary. The model vocabulary is built using a large text corpus, to which an integer index ID is assigned for each frequent token.
- **Input IDs:** The token indices obtained in the previous step are numerical IDs. The sequence of numerical IDs is also known as input IDs, which form the input of the BERT model.
- **Padding & Truncation:** are performed to ensure all input sequences are uniform in length. Larger input sequences are truncated to a specified maximum length. The [PAD] special token is used for padding sequences to a fixed length. This helps to ensure that the input sequences have the same length, which helps the BERT model achieve computational efficiency. Generally, [PAD] tokens do not contribute any meaning to the input sequence; they only help the model distinguish between actual and padded tokens.
- **Attention Masks:** ensure that the model only focuses on important parts of the input sequence. The input sequence also contains padding tokens to prevent the model from focusing; thus, padding token attention marks are introduced. These attention masks are binary matrices. Padding tokens are given a 0 value, and all other important tokens are given a value of 1.

3.4. BERT fine-tuning

The BERT model is fine-tuned to adapt its pre-trained weights specifically for ABSA tasks. Instead of default settings, the optimal hyperparameter values were determined through systematic experimentation. To achieve this, an ablation study was conducted rather than relying solely on default settings. An ablation study is a systematic approach to assess how hyperparameters affect model performance. During the fine-tuning process, the BERT model parameters are adjusted through back-propagation. This helps the BERT model to capture task-specific patterns from the AWARE dataset. The ablation study in Section 4.2 offers a thorough analysis of the effect of important hyperparameters on the model performance. Table 2 shows the configuration of the best hyperparameters used for fine-tuning the BERT model. As per the above hyperparameter configuration, each fine-tuning run is designed based on 16 epochs. Due to GPU memory limitations, the BERT model is trained with a batch size of 8. A smaller batch size with a smaller learning rate of 1e−5 empirically proved to be more effective because it helps the model generalize better. In addition, the training of the model is observed to be more stable with this learning rate. During the fine-tuning process, the BERT model is compiled with an Adaptive Moment Estimation (ADAM) optimizer (Kingma and Ba, 2014). The ADAM optimizer is the best choice, as suggested by the majority of literature (Gkouti et al., 2024) because of its adaptive learning rate and efficient convergence properties. Sparse categorical cross-entropy is suitable for multi-class classification tasks such as ABSA. In addition, a sparse categorical cross-entropy loss function was used to measure the difference between the predicted and actual class labels.

Table 2
Best hyperparameters for BERT fine-tuning.

Parameter	Value
Batch Size	8
Learning Rate	1e−5
Epochs	16
Optimizer	Adam
Max Seq Length	256
Dropout Rate	0.2
Activation	Softmax
Loss	Sparse categorical cross entropy

3.5. Model construction

Our fine-tuned BERT model is constructed using the $\text{BERT}_{\text{base}}$ model. The internal architecture of our BERT model is shown in Fig. 3. The BERT model uses a review sentence as input. These review sentences are tokenized into smaller units called tokens. Special tokens, such as the [CLS] token for the classification task and [SEP] token for the separation of input sequences are added to prepare the input sequence for the BERT model. The prepared input sequence is passed to the BERT model. The BERT model efficiently uses the encoder part of transformer architecture to capture rich contextual information from a given input sequence. The $\text{BERT}_{\text{base}}$ model consisted of 12 encoder layers in the transformer architecture. Each encoder layer in the BERT model performs the following tasks:

1. Multi-Head Self-Attention Mechanism: This is the main component of transformer architecture, which helps classification models to analyze input sequences from various perspectives. This enables a contextually informed and comprehensive representation of the text. The process starts with the computation of self-attention matrix A, which is derived from a given input sequence $X = \{x_1, x_2, \ldots, x_n\}$. The input sequence X is a collection of token embeddings, where each token embedding x_i represents the corresponding word or sub-word of the input sequence in its vector representation form. Each entry in the attention matrix x_{ij} shows the relevance and importance of token x_i with respect to token x_j.
   The input sequence X is first converted into Query (Q), Key (K), and Value (V) matrices in order to compute the self-attention matrix A. The concept of Query (Q) can be considered as a representation of the current item the model is focusing on in an input sequence. On the other hand, the Key (K) represents all the items the model can potentially focus on. Each Key (K) is associated with the Value (V). Once the model identifies which keys are important and relative to the query, the model retrieves the corresponding values. These values are then used to construct the output representations from the input sequence. From Q and K the attention score is calculated using a scaled dot-product operation, as given in Eq. (1):
   $$\text{Attention}(Q, K) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \tag{1}$$
   where d_k denotes the dimensionality of the key vectors. Finally, multiplying the attention score given in Eq. (1) with the value

matrix yields the self-attention matrix, which is mathematically given as

$$A = \text{Attention}(Q, K) \cdot V \tag{2}$$

BERT benefits from a multi-head attention mechanism in which each head refers to a separate self-attention system that works concurrently. Mathematically, multi-head attention is expressed by Eq. (3).

$$\text{MultiHead}(Q, K, V) = \text{Concat}(h_1, h_2, \dots, h_h)W_O \tag{3}$$

where h_i represents a separate attention head and W_O is the final learned weight matrix for the final output. As we used the $\text{BERT}_{\text{base}}$ model, it utilizes 12 layers of transformer encoder architecture, each with 12 attention heads. This configuration led to a total of $12 \times 12 = 144$ distinct attention mechanisms within the model. Multiple heads in BERT process multiple aspects of the input sequences in parallel, which helps BERT better understand the contextual representation of a given input sequence. Additionally, considering various aspects of the input sequence makes BERT capable of generalizing across various tasks.

2. Residual Connections and Layer Normalization: A residual connection is introduced to integrate the output from the multi-head self-attention mechanism with the original input sequence. This step preserves important information by capturing complex patterns and dependencies. In this step, the BERT model considers learning from both the original input sequences and the attention-based representations. This step is further followed by layer normalization, which helps the BERT model stabilize the training process by promoting stable activation within each layer of the network, resulting in smoother convergence during training.
3. Feed-Forward Neural Network: The normalized output from the previous layer underwent further processing using a Feed-Forward Neural Network (FFNN). This network configuration contains two fully connected layers, and a nonlinear activation function Gaussian Error Linear Unit (GELU) is applied between them. In this step, the model learns to capture context-rich representations and meaningful features within the input sequence, preparing it for downstream tasks, such as text classification or question answering. After processing from the FFNN, another round of residual connection and layer normalization occurs, in which the output from the FFNN is combined with the original input to the FFNN, followed by layer normalization. This step helps preserve important information by ensuring that the model learns from the original input sequence and previous layers, as well as from newly learned features using the FFNN. Potential issues such as vanishing gradients are mitigated using residual connections and layer normalization.

The output from each encoder layer in BERT serves as input to the next encoder layer in a sequential manner, creating a stack of a total of 12 encoders that build upon each other in $\text{BERT}_{\text{base}}$ model. Together, the above-mentioned sequence of operations performed for all encoder layers in BERT contributes to the enhanced learning process and robustness of the BERT model performance. The [CLS] token is a special token added at the beginning of the input sequence for the classification task. This provides a summary of the entire input sequence. The [CLS] token, in conjunction with the input sequence, is also passed through the stack of encoder layers where attention is computed, and the input sequence is processed by the FFNN. The embedding of the [CLS] token helps capture the relationships within the input sequence and overall context. When the processing of the input sequence reaches the final encoder layer in BERT, the embedding of the [CLS] token represents the global context, dependencies, and relationships across the input sequence. For downstream tasks, such as classification, the embeddings of [CLS] serve as inputs for further processing.

We introduce an FFNN for the ABSA classification task in the $\text{BERT}_{\text{base}}$ model after the [CLS] token. The embedding of the [CLS] tokens is passed through a dense layer in FFNN for the ABSA tasks. This approach is also suggested for various classification tasks by Sun et al. (2019) and Devlin et al. (2018). The output from the FFNN dense layer is passed after processing to the softmax layer, which helps convert the model's output probability distribution into different classes. The class with the highest probability is considered the class predicted by the softmax layer for ABSA tasks.

The fine-tuned BERT model with the best hyperparameters serves as a foundation for the construction of a classification model that can perform ACP and ACD tasks. For this purpose, we first need to prepare a dataset. For model dataset preparation, the pre-processed the AWARE dataset was utilized. To ensure a consistent and accurate evaluation of the experimental results, we used the same dataset splitting strategy as specified in the baseline study (Alturayeif et al., 2023).

To maintain an equal distribution of classes across the different splits, we used a 10-fold Stratified K-Fold Cross-Validation technique. The dataset was divided into ten folds, and each fold was used as the test set. The remaining nine partitions (90%) of the data were used for the training and validation. The training data (81% of all samples) were used to provide model generalization, and the remaining data (approximately 9% of the entire dataset) were used as validation data to enable the performance assessment of the model during the training phase on unseen data. Moreover, the use of the test set is dynamic where 10% of the dataset is used from each of the fold, such that every data point is used once as a test sample thus eliminating the bias and giving a fair evaluation of the model and its ability to generalize.

3.6. XAI model

After successfully fine-tuning the BERT model for the ABSA tasks, a classification model was constructed, which helped achieve outstanding prediction results for both ACD and ACP tasks. However, classification models are black-box in nature, owing to their lack of transparency and interpretability. To address this issue, we integrated XAI into our fine-tuned BERT classification model. In this approach, we use our classification model and create an XAI model to obtain better interpretations of the model's predictions in a transparent manner. We selected LIME an XAI technique to add explainability to our classification model and the ABSA results. This section discusses in detail how we integrated XAI and explains the model prediction results.

3.6.1. LIME

LIME is a model-independent interpretability approach for local individual predictions. Interpretability refers to the degree to which a person can understand why a particular prediction choice is made using a black-box model. The authors of the LIME model argue that it is necessary to establish two types of trust for a user to adopt a model: (1) trusting a prediction because the users are not interested in blindly trusting model predictions, and (2) trusting a model to ensure that as the user gains enough trust, the model will behave rationally when deployed. The LIME model helps establish this trust, which fosters transparency and confidence in using black-box models. The results from the BERT model are further applied to the LIME model for local explanation.

4. Results & discussion

Our study attempted to transform the traditional BERT model for ABSA tasks using BERT fine-tuning. We employed a unique approach based on XAI to render the BERT classification model interpretable and explainable. This section discusses in detail the results of experiments performed to fine-tune BERT for ABSA and the integration of XAI in it for post-hoc explanations of model predictions.

Table 3
Baseline configuration.

Study Ref.	Model	Configuration
Alturaief et al. (2021)	SVM	SVM with TF-IDF (Term Frequency-Inverse Document Frequency) Features
Alturayeif et al. (2023)	BERT	Batch size: [16, 32], Learning rate: [5e−5, 3e−5, 2e−5] Epochs: [2, 3, 4], Optimizer: Adam
Gunathilaka and De Silva (2022)	CNN	word embedding: Word2Vec, Learning Rate: 0.001, Epochs: 75, Batch Size: [25, 35], Dense Activation: Sigmoid, CNN Activation: ReLU
Our Model	BERT (Fine-tuning)	Batch size: 8, Learning rate: 1e−5, Epochs: 16, Optimizer: Adam

4.1. Experimental results

In this section, the experimental results are discussed in detail. The source code used in the experimental design is made public at Github.[4]

In order to evaluate the proposed method, we compare its results with the baseline models from existing research. These baselines include conventional machine learning models and deep learning models like SVM, CNN, and BERT-based models. Table 3 contains the model description and configuration setups for these models. Accuracy, precision, recall, F1-Score, and confusion matrix were used as the performance metrics to evaluate the performance of the models.

Our proposed fine-tuned BERT model achieved SOTA results and outperformed existing baseline results. The baseline model based on SVM by Alturaief et al. (2021) achieved an F1-Score of 0.32 for the AWARE Social Networking & Games domain and 0.33 for the AWARE Productivity domain on the ACD task. For the ACP task, their SVM model achieved an accuracy of 0.69 for the AWARE Social Networking domain, 0.68 for the AWARE Productivity domain, and 0.67 for the AWARE Games domain. Alturaief et al. extended their work in Alturayeif et al. (2023) and achieved comparatively good results by using the BERT model. Their model achieved an ACD task F1-Score of 0.43 for the AWARE Social Networking domain, 0.42 for the AWARE Productivity domain, and 0.47 for the AWARE Games domain. An accuracy of 0.81 is obtained for the Social Networking domain and 0.77 for the AWARE Productivity and AWARE Games domains. Another study conducted in Gunathilaka and De Silva (2022) used the AWARE dataset and proposed a model based on a Convolutional Neural Network (CNN) architecture. The authors concluded that CNN, along with word2Vec, performed well and achieved an F1-Score of 0.62 for the AWARE Social Networking and Productivity domain and 0.42 for the AWARE Games domain. For the ACP task, their model achieved accuracies of 0.86, 0.80, and 0.70 for the AWARE Social Networking, Productivity, and Games domains, respectively. In this section, we discuss the results of our proposed model in detail and compare our results with the baselines on different subtasks related to ABSA.

4.1.1. ACD task results

This section discusses the ACD task results obtained using the proposed fine-tuned BERT model. To offer a comprehensive overview of model performance across all three domains of the AWARE dataset, the results are organized domain-wise.

1. ACD Task Results for the AWARE Social Networking Domain: For the AWARE Social Networking domain, our fine-tuned BERT model outperformed existing models, as shown in Table 4. For better visualization of the model performance, the confusion matrix is illustrated in Fig. 4(a) and Table 5 shows the classification report of the ACD task for the AWARE Social Networking domain.

[4] https://github.com/engrsoonhtaj/BERT-for-RE

Table 4
Comparison of F1-scores ACD task: Social networking domain.

Study	Approach	F1-Score
Alturaief et al. (2021)	SVM	0.32
Alturayeif et al. (2023)	BERT	0.43
Gunathilaka and De Silva (2022)	CNN	0.62
Our Work	BERT (Fine-tuning)	0.78

Table 5
AWARE social networking domain: Classification report across folds for ACD task.

	Precision	Recall	F1-Score	Support
Cost	0.95	0.93	0.94	195
Effectiveness	0.74	0.69	0.71	385
Efficiency	0.60	0.72	0.65	339
Enjoyability	0.71	0.69	0.70	94
General	0.83	0.80	0.82	757
Reliability	0.73	0.73	0.73	180
Safety	0.80	0.71	0.75	34
Security	0.92	0.90	0.91	289
Usability	0.78	0.78	0.78	845
Accuracy			0.78	3118
Macro Avg	0.79	0.77	0.78	3118
Weighted Avg	0.79	0.78	0.78	3118

Table 6
Comparison of F1-scores ACD task: Productivity domain.

Study	Approach	F1-Score
Alturaief et al. (2021)	SVM	0.33
Alturayeif et al. (2023)	BERT	0.42
Gunathilaka and De Silva (2022)	CNN	0.62
Our work	BERT (Fine-tuning)	0.82

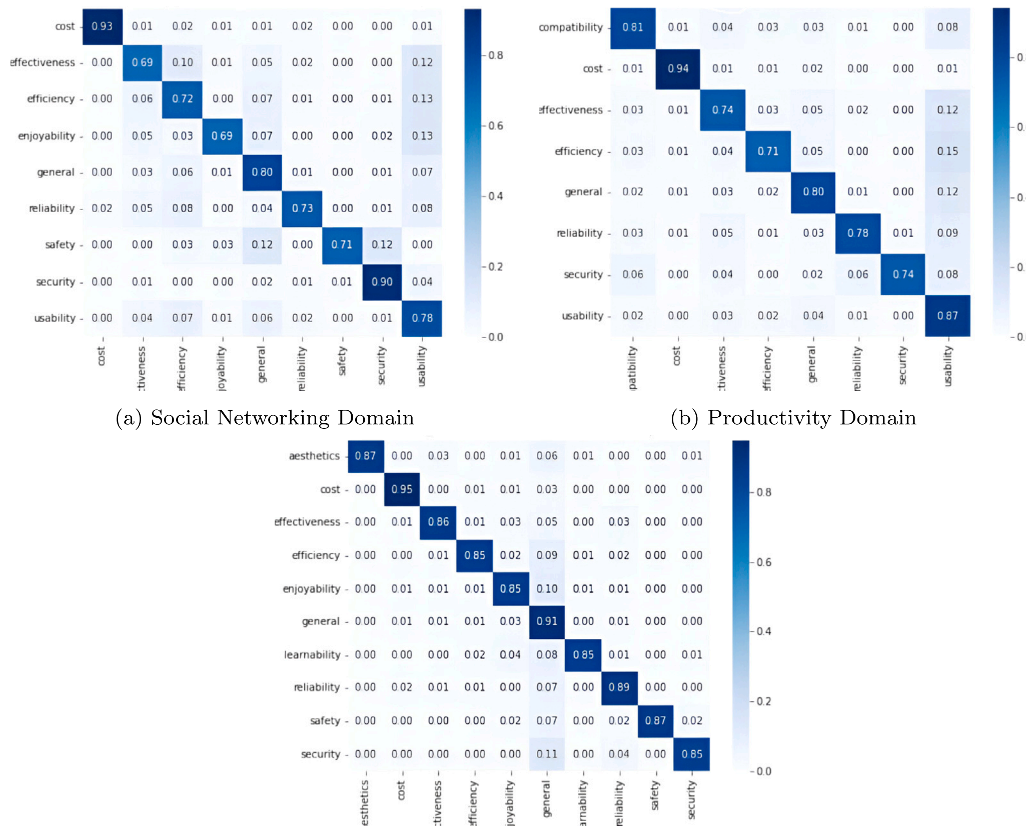
Table 7
AWARE productivity domain: Classification report across folds for acd task.

	Precision	Recall	F1-Score	Support
Compatibility	0.78	0.81	0.80	357
Cost	0.91	0.94	0.93	297
Effectiveness	0.79	0.69	0.71	520
Efficiency	0.78	0.72	0.65	325
General	0.83	0.80	0.82	751
Reliability	0.80	0.78	0.79	164
Security	0.84	0.74	0.79	50
Usability	0.82	0.87	0.84	1310
Accuracy			0.82	3774
Macro Avg	0.82	0.80	0.81	3774
Weighted Avg	0.82	0.82	0.82	3774

2. ACD Task Results for the AWARE Productivity Domain: Similar to the AWARE Social Networking domain, our fine-tuned model performed well in the Productivity domain and outperformed the existing benchmark results. Comparison is given in Table 6. To better understand the model performance, a confusion matrix is visualized in Fig. 4(b). Table 7 presents the classification report for the ACD task in the AWARE Productivity domain.
3. ACD Task Results for the AWARE Games Domain: Our fine-tuned BERT model also proved to perform best for ACD tasks on the AWARE Games domain. A comparison with existing literature is presented in Table 8. In addition, to visualize the model performance, the confusion matrix is shown in Fig. 4(c). Table 9 also offers a visualization of the classification report for the ACD task on the AWARE Games domain.

4.1.2. ACP task results

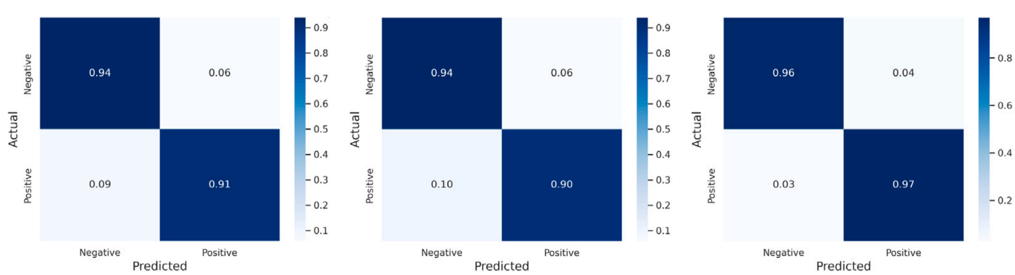
1. ACP Task Results for the AWARE Social Networking Domain: For the AWARE Social Networking domain, our fine-tuned BERT model achieved the highest accuracy score compared to existing



(a) Social Networking Domain (b) Productivity Domain

(c) Games Domain

Fig. 4. Confusion matrix (percentages) ACD task.



(a) Social Networking Domain (b) Productivity Domain (c) Games Domain

Fig. 5. Confusion matrix (percentages) of ACP task.

Table 8
Comparison of F1-scores ACD task: Games domain.

Study	Approach	F1-Score
Alturaief et al. (2021)	SVM	0.32
Alturayeif et al. (2023)	BERT	0.47
Gunathilaka and De Silva (2022)	CNN	0.42
Our work	BERT (Fine-tuning)	0.89

Table 9
AWARE games domain: Classification report across folds for ACD task.

	Precision	Recall	F1-Score	Support
Aesthetics	0.93	0.87	0.90	71
Cost	0.93	0.95	0.94	346
Effectiveness	0.88	0.86	0.87	281
Efficiency	0.74	0.86	0.80	254
Enjoyability	0.90	0.87	0.88	815
General	0.90	0.90	0.90	1461
Learnability	0.83	0.83	0.83	127
Reliability	0.87	0.86	0.87	255
Safety	0.97	0.83	0.89	46
Security	0.80	0.85	0.83	53
Accuracy			0.89	3709
Macro Avg	0.88	0.87	0.87	3709
Weighted Avg	0.89	0.89	0.89	3709

works. A comparison is presented in Table 10. To provide an in-depth view of the model performance, we have added the confusion matrix and classification report here, as shown in Fig. 5(a) and Table 11.

2. ACP Task Results for the AWARE Productivity Domain: Similar to the AWARE Social Networking domain, our fine-tuned BERT model provided superior accuracy results for the AWARE Productivity domain. It achieved the highest accuracy compared to the existing methods found in the literature. For a better

Table 10
Comparison of accuracies for ACP Task: Social networking domain.

Study	Approach	Accuracy
Alturaief et al. (2021)	SVM	0.69
Alturayeif et al. (2023)	BERT	0.81
Gunathilaka and De Silva (2022)	CNN	0.86
Our work	BERT (Fine-tuning)	0.92

Table 11
AWARE social networking: Classification report across folds for ACP task.

	Precision	Recall	F1-Score	Support
Negative	0.92	0.94	0.93	1724
Positive	0.92	0.91	0.91	1394
Accuracy			0.92	3118
Macro Avg	0.92	0.92	0.92	3118
Weighted Avg	0.92	0.92	0.92	3118

Table 12
Comparison of accuracies for ACP task: Productivity domain.

Study	Approach	Accuracy
Alturaief et al. (2021)	SVM	0.68
Alturayeif et al. (2023)	BERT	0.77
Gunathilaka and De Silva (2022)	CNN	0.80
Our work	BERT (Fine-tuning)	0.92

Table 13
AWARE productivity domain: Classification report across folds for ACP task.

	Precision	Recall	F1-Score	Support
Negative	0.90	0.94	0.92	1801
Positive	0.94	0.90	0.92	1973
Accuracy			0.92	3774
Macro Avg	0.92	0.92	0.92	3774
Weighted Avg	0.92	0.92	0.92	3774

Table 14
Comparison of accuracies for ACP task: Games domain.

Study	Approach	Accuracy
Alturaief et al. (2021)	SVM	0.67
Alturayeif et al. (2023)	BERT	0.77
Gunathilaka and De Silva (2022)	CNN	0.70
Our work	BERT (Fine-tuning)	0.96

Table 15
AWARE games: Classification report across folds for ACP task.

	Precision	Recall	F1-Score	Support
Negative	0.96	0.96	0.96	1766
Positive	0.96	0.97	0.97	1943
Accuracy			0.96	3709
Macro Avg	0.96	0.96	0.94	3709
Weighted Avg	0.96	0.96	0.94	3709

understanding, a comparison of the performance of our fine-tuned BERT model and existing methods is presented in Table 12. In addition to this comparison, the comprehensive insights about our fine-tuned BERT models predictive capabilities are given in the confusion matrix and classification report, as shown in Fig. 5(b) and Table 13.

3. ACP Task Results for the AWARE Games Domain: Our fine-tuned BERT model outperformed the existing methods, and a comparison is presented in Table 14. Furthermore, to provide better insights into model performance, we generated a confusion matrix and a classification report, as shown in Fig. 5(c) and Table 15.

Table 16
Review samples for ACD task.

Review #	Review sample text
1	"*Update* I changed the rating to two stars because now the search function does not work like it used to."
2	"So, in the recent updates, Facebook finally synced to dark mode, but now that feature is gone?"
3	"It is unfair that persons have lost their accounts so suddenly and are not getting the proper help they so deserve."
4	"It is fun, it is challenging, it is a murder mystery."
5	"It makes it feel like a mobile game where there are ways to spend money around every corner."

Table 17
Evaluation results of predicted classes ACD task.

Review #	Predicted class	Local fidelity	Perturbation stability
1	Efficiency	0.762128	0.775382
2	Reliability	0.930573	0.800153
3	Security	0.846749	0.860963
4	Enjoyability	0.990878	0.983807
5	Cost	0.999723	0.991400

4.1.3. XAI results: LIME

Two metrics were employed to measure the performance of the LIME model: the Local Fidelity and Perturbation Stability.

- Local Fidelity: It is a metric used to see the correlation between BERT model classification results and explanations produced by LIME. This metric helps us to quantify how closely the explanations produced by the LIME model match the actual classification results. A higher value of local fidelity denotes a stronger alignment between the model prediction and the explanations provided by LIME.
- Perturbation Stability: This metric measures the robustness and consistency of the classification model by exposing its input data with small changes or perturbations. This metric also provides robustness to the explanations produced by the LIME model. This is because small variations in the input data do not affect the models prediction; therefore, the explanations provided by LIME are also likely to be more stable when subjected to variations in the input data.

1. ACD Task LIME Results: This section discusses the ACD task results obtained using the LIME. The LIME XAI method was applied as a post-hoc explanation to evaluate and interpret the BERT model classification results. For this purpose, we randomly selected five samples from the AWARE dataset, as shown in Table 16.
   For each sample, LIME provides highlighted text, visual explanations, and local explanations. One of the LIME explanations for these samples is shown in Fig. 6.
   Furthermore, for each randomly selected review sample, the Local Fidelity and Perturbation Stability scores were also calculated to obtain a comprehensive idea of the models performance using the LIME explanation method. Table 17 lists the Local Fidelity and Perturbation Stability scores. It can be observed that for each review sample, LIME provided correctly predicted classes, and both scores were high, which contributed to the robustness of both the LIME explanations and BERT model classification results.
2. ACP Task LIME Results: For the ACP task, we also evaluated the BERT model prediction results using the LIME explanations; for this, we took 5 random samples from the AWARE dataset, also given in Table 18.
   LIME provides highlighted text, visual explanations, and local explanations for the review samples. A LIME explanation for one of these samples is shown in Fig. 7.

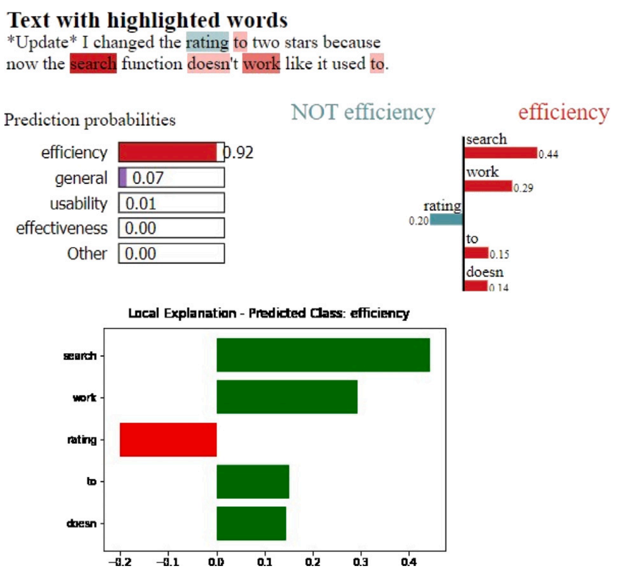


Fig. 6. ACD task: LIME explanation with highlighted review text, visual explanation, and local explanation for review sample 1.

Table 18
Review samples ACP task.

Review #	Review sample text
1	"This feels like an invasion of privacy and a scam to get more data to sale."
2	"So much fun and definitely recommend it!"
3	"There seems to be multiple bugs after the most recent update."
4	"I love the layout and user friendly interface."
5	"I am one of the lucky users to be affected by your alleged server issues."

Table 19
Evaluation results of predicted classes ACP task.

Review #	Predicted class	Local fidelity	Perturbation stability
1	Negative	0.993284	0.929236
2	Positive	0.999998	0.999998
3	Negative	0.999442	0.998457
4	Positive	0.999989	0.999997
5	Negative	0.984478	0.931424

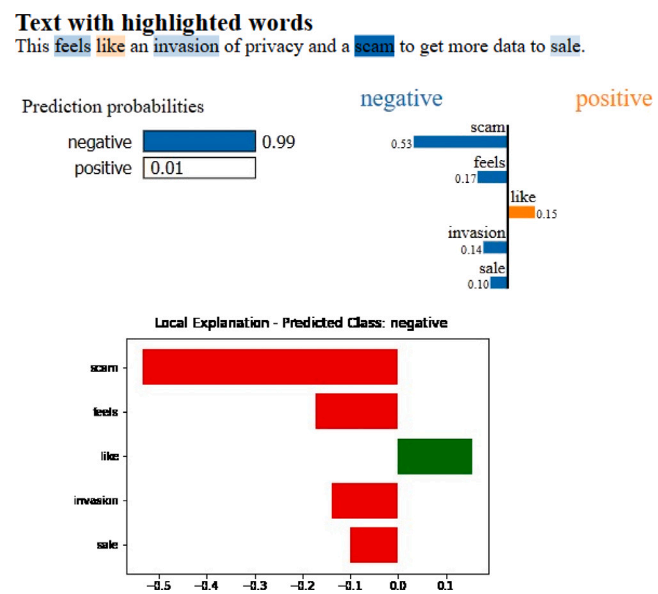


Fig. 7. ACP task: LIME explanation with highlighted review text, visual explanation, and local explanation for review sample 1.

In addition, for each random review sample, the Local Fidelity and Perturbation Stability scores were also calculated, which helped evaluate both the BERT classification model results and LIME explanations. Table 19 shows the Local Fidelity and Perturbation Stability scores for the selected five random review samples, where it can be clearly seen that for each review sample, LIME provided accurately predicted classes with a high score of Local Fidelity and Perturbation Stability. These results help assess the trustworthiness of LIME explanations and the stability of the BERT classification results for the ACP task.

Though LIME shows certain strengths regarding local fidelity and perturbation stability for the interpretation of BERT predictions within this study, some other limitations persist. Here is a brief discussion of those limitations.

(a) **Local Fidelity vs. Global Understanding:** In LIME, explanations are generated through a local approximation of the decision boundary, which includes an instance of interest. As BERT is a model with such high complexity, local approximations can deviate from the global behavior of the model so these local approximations can provide misleading explanations.

(b) **High Computational Cost:** LIME explanation generation involves the use of surrogate models for point-level analysis, accompanied by the creation of multiple perturbations. The process may also be computationally expensive and time-consuming especially when dealing with large datasets and complex models, such as BERT.

(c) **Lack of Intrinsic Insights:** LIME is defined as a model-agnostic technique, which means that the method does not use the unique features of BERT, such as attention weights or hidden representations, to explain the model's decisions. To accomplish this, a local surrogate that approximated the behavior of the model at different points was used. Consequently, LIME-assigned explanations might not properly describe the intrinsic inner workings and the decision-making process of BERT while accepting inputs, which could further lead to misinterpretation of BERT's thinking process, resulting in a disconnection between the explanation and the real functioning of the model.

(d) **Tokenization Misalignment and Lack of Stability in Explanations:** In contrast to LIME, which typically works with a word or token level tokens, BERT subword tokenizes words with subword tokenization, like WordPiece or Byte Pair Encoding. Such a mismatch can make it difficult to ensure that LIME's explanations are consistent with the BERT's actual input tokens. In addition, LIME's explanations are created with randomness in the sense of generating perturbations, which may in turn affect the reliability and stability of its explanations.

4.2. Ablation study

When fine-tuning BERT, hyperparameters are crucial and significantly impact the performance of the model. Parameters such as the learning rate, batch size, number of epochs, dropout rate, and weight decay influence model training and generalization to new data. In this study, we performed several experiments to observe the effects of hyperparameter tuning on the performance of the BERT model. We investigated the impact of the number of epochs, batch size, learning rate, and choice of optimizer on the model performance. This section

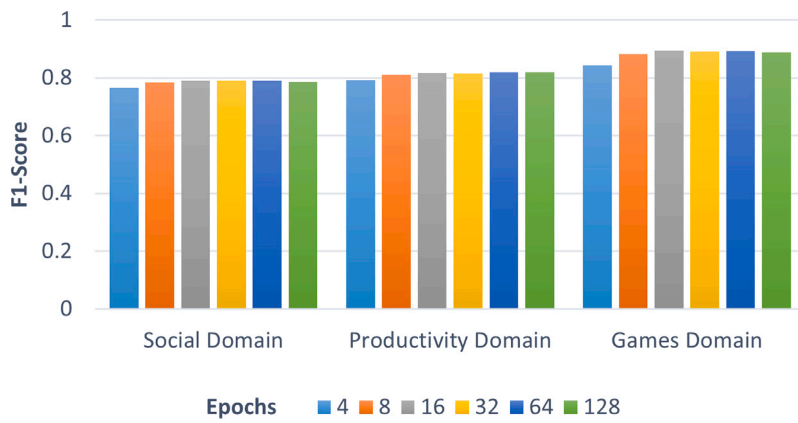


Fig. 8. ACD task: F1-Score for various epochs.

specifically discusses the impact of hyperparameters on the fine-tuning of BERT.

4.2.1. Impact of number of epochs on BERT fine-tuning

The first round of experiments was conducted to assess the effect of the number of epochs on the performance of the BERT model. Experiments were conducted for three domains of the AWARE dataset: Social Networking, Productivity, and Games. Results and analysis for both tasks, i.e., ACD and ACP tasks performed with different numbers of epochs to demonstrate BERT model performance are illustrated in Figs. 8 and 9.

Effect of the number of epochs on ACD task: Our fine-tuned BERT model without any resampling technique outperformed existing baseline studies. For the ACD task, the F1-Score was considered a suitable metric for observing the model's performance with varying epochs for all domains. The F1-Score considers both precision and recall, which are helpful for the class imbalance issues observed in aspect categories. The results of the ACD task are summarized below.

1. AWARE Social Networking Domain: For the AWARE Social Networking domain, the F1-Score ranged from 0.7646 to 0.7903 across different epoch values. Demonstrate a gradual increase up to 16 epochs. After this point, the F1-Score slightly declined.
2. AWARE Productivity Domain: Similar trends are observed for the AWARE Productivity domain, with the F1-Score ranging from 0.7924 to 0.8182. The peak F1-Score was observed at 16 epochs; after 16 epochs, there was no substantial increase in the F1-Score.
3. AWARE Games Domain: For the AWARE Games domain, the F1-Score also showed significant improvement up to 16 epochs, reaching the highest value of 0.8936. After this point, the F1-Score slightly declined with additional epochs.

Effect of number of epochs on ACP task:

1. AWARE Social Networking Domain: As the number of epochs increased, model performance gradually increased. The validation accuracy peaked at 16 epochs, with a validation accuracy of 0.92. However, it was observed that the model performance decreased when the number of epochs increased. Therefore, we conclude that the model achieved the highest validation accuracy at 16 epochs. However, the training accuracy increased with increasing epochs, which demonstrates the model's ability to learn from the training data. It was also observed that with a range of epochs, the test accuracy remained stable.
2. AWARE Productivity Domain: Likewise, in the Social Networking domain, the model achieved an optimum validation accuracy of 0.93 at 16 epochs. The performance of the model gradually decreased with further epochs. Training accuracy consistently increased as the number of epochs increased. This demonstrates the effectiveness of the model in learning from the training data. The test accuracy appeared to be relatively stable, ranging between 0.9259 and 0.9296 across different numbers of epochs.
3. AWARE Games Domain: The model exhibits performance improvements and reached a peak validation accuracy of 0.95 at 16 epochs. However, the validation accuracy decreases as the number of epochs increases. This indicates a decrease in the model performance after 16 epochs. For the other domains, the training accuracy slightly increased with the number of epochs. This indicates the ability of the model to learn from training data. The test accuracy remained relatively constant, ranging from 0.9571 to 0.9647 over various epochs.

In conclusion, for all the domains, the training accuracy increased with an increase in the number of epochs. However, the impact on the test and validation accuracies differed for the various epochs. The model achieved optimal performance at 16 epochs, with the highest validation accuracy for all domains. Additionally, the model performance declined with an increasing number of epochs beyond this point. The training accuracy was stable across different numbers of epochs for all the domains. The above results suggest the importance of choosing the right number of epochs to achieve the best performance results on unseen data while preventing models from overfitting.

These results suggest that 16 epochs appear to be a practical choice for all the AWARE dataset domains and both tasks, that is, ACD and ACP, especially when computational resources are concerned. The above experiments also provide evidence that beyond 16 epochs, the model's performance declined due to over-fitting; therefore, it can be concluded that 16 epochs provided the model's peak performance by creating a balance between training accuracy and generalization to unseen data.

4.2.2. Impact of batch size selection on BERT fine-tuning

For the ACD task, a batch size of 8 yielded a high F1-Score for all the AWARE dataset domains compared to a larger bath size of 16 and 32. The same trend was observed for the ACP task. Employing a smaller batch size of eight resulted in high validation accuracy compared to larger batch sizes, such as 16 or 32, for all AWARE dataset domains. These observations are shown in Figs. 10(a) and 10(b), which clearly show that fine-tuning smaller batch sizes is more suitable than fine-tuning larger batch sizes for BERT. The literature also supports this observation, as many factors, such as smaller batch sizes, help the model learn important features and enhance its performance and generalization capability for unseen data (Sun et al., 2019). Smaller batch sizes also contributed to the smooth convergence of the model during the training phase.

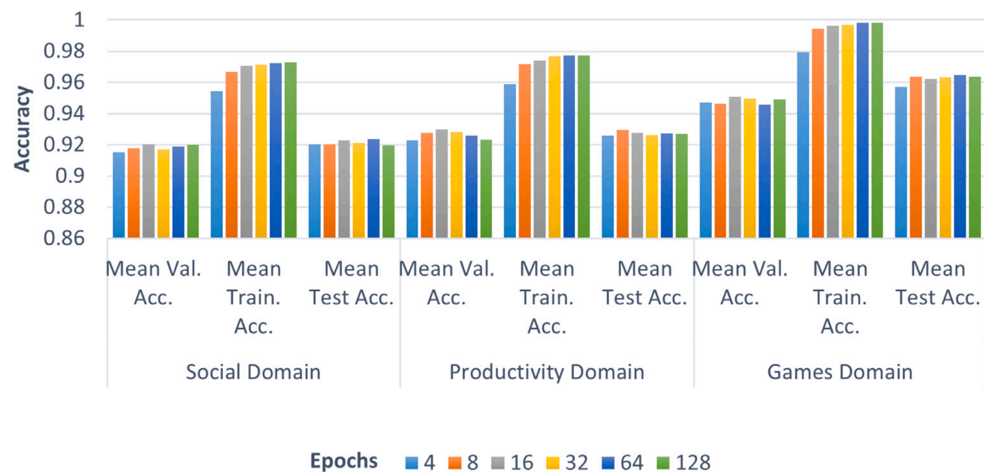
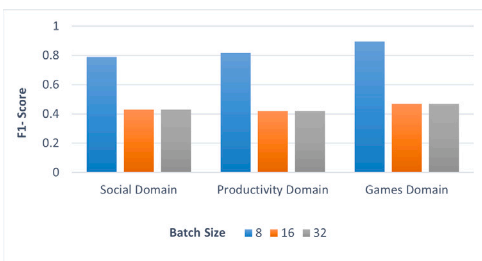
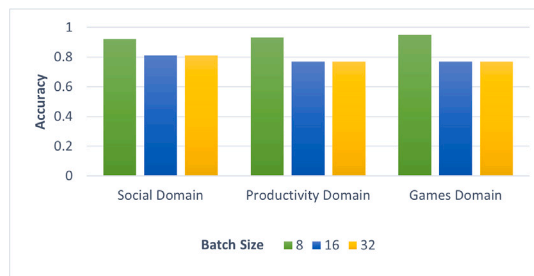


Fig. 9. ACP task: Accuracy score for various epochs.

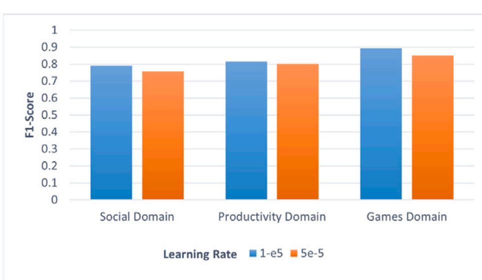


(a) ACD Task Batch Size

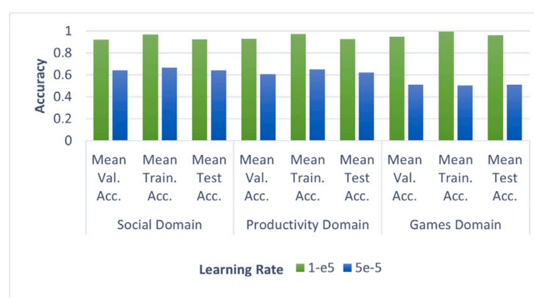


(b) ACP Task Batch Size

Fig. 10. Impact of batch size selection on BERT fine-tuning.



(a) ACD Task Learning Rate



(b) ACP Task Learning Rate

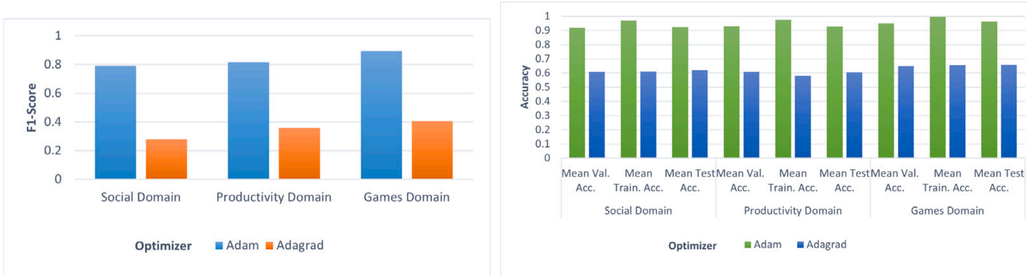
Fig. 11. Impact of learning rate on BERT fine-tuning.

4.2.3. *Impact of learning rate on BERT fine-tuning*

Learning rate is one of the most important hyperparameters in BERT fine-tuning. The learning rate choice determines the step size of the optimization algorithm when updating the model parameters during the training phase. The learning rate also affects the model convergence and stability during the training phase. Therefore, choosing a suitable learning rate is important for fine-tuning BERT. In this section, the results for different learning rates are presented, and for the experiments, we used one low learning rate and one high learning rate, that is, 1e−5 and 5e−5, respectively. From the graph presented in Fig. 11(a), it is evident that for the ACD task, the low learning rate outperformed the high learning rate by providing a high F1-Score across all the AWARE dataset domains. The same happened for the ACP task; a low learning rate provided the best performance results across all domains compared to a higher learning rate, as shown in Fig. 11(b). These results suggest that a low learning rate is better for fine-tuning BERT. This could be due to several reasons. For instance, low learning rates prevent the model from overfitting, with slower convergence. By contrast, a high learning rate converges rapidly and causes model overfitting. For this reason, during BERT fine-tuning, a higher learning rate does not provide good model performance results.

4.2.4. *Impact of optimizer selection on BERT fine-tuning*

The performance of the BERT model, including its convergence speed and stability in handling unseen data, depends on the selection of the optimizer. Optimizers such as AdaGrad, ADAM, and their variants are commonly used for various NLP tasks, each with its own strategy to update model parameters during training. AdaGrad updates its learning



(a) ACD Task Optimizer (b) ACP Task Optimizer

Fig. 12. Impact of optimizer selection on BERT fine-tuning.

rate for each parameter based on the historical gradients gathered during the training. AdaGrad is effective at handling sparse data commonly encountered in text datasets. The ADAM optimizer is more robust in terms of performance and provides a faster convergence rate because it incorporates the momentum and adaptive learning rates of each parameter by considering the mean and variance of the gradients. ADAM combines the features of both the AdaGrad and RMSProp optimizers. The summarized results for the performance of both optimizers when fine-tuning the BERT model are shown in Figs. 12(a) and 12(b). ADAM demonstrated superior performance for ACD and ACP tasks compared to AdaGrad in terms of F1-Score metrics.

Instead of relying on default values, the hyperparameter configuration in Table 2 is the result of a systematic ablation study. The ablation study discussed above consisted of choosing an optimal configuration by evaluating the impact of different hyperparameters on the model's performance. In particular, it was discovered that a learning rate of 1e−5 and a batch size of 8 combined with 16 training epochs using the Adam optimizer provide the best results. Moreover, we set the maximum sequence length to 256 and the dropout rate of 0.2 to prevent overfitting. Softmax activation function and sparse categorical cross entropy loss function were used to make the classification more effective. Each parameter was systematically analyzed and optimized to maximize BERT's capacity to learn task-specific patterns. This means that the chosen hyperparameters are justified by empirical results instead of arbitrary choices.

5. Limitations and future work

Our research laid the foundation for improved requirement elicitation through Aspect-Based Sentiment Analysis using BERT fine-tuning and XAI integration. However, there are some limitations of our work that need to be addressed in order to further improve the proposed approach's robustness and applicability.

1. **Limited Datasets Exploration and Domains Coverage:** The model performs well on the AWARE dataset, yet the model has not been tested on larger-scale datasets other than the AWARE dataset. From this, one can consider scaling the model to larger datasets may need larger computational resources and longer training times. Our proposed model has been evaluated exclusively on the AWARE dataset, which focuses on app reviews within three specific domains: Productivity, Social Networking, and Games. These domains promise a sufficient variety and diversity. However, the evaluation of this approach was limited to a single AWARE dataset with limited application domains. Actual RE tasks imply working with datasets originating from different domains, such as healthcare, finance, or e-commerce, which might have different languages, different sets of vocabulary, and data distributions. This limits the degree to which we can generalize our findings to other RE datasets underpinning other domains.
2. **Limited Cross-Lingual Generalizability and Handling of Noisy App Reviews:** The current work only focuses on English language data, making it hard to use for multilingual and cross-lingual Requirement Engineering (RE) tasks. To grasp the requirements, RE processes usually involve officially documented and textual sources, such as user reviews, app feedback, and interviews. In globalized industries, these inputs may exist in multiple forms, including low-resource languages that lack labeled datasets. However, the performance of our proposed approach in these multilingual or low-resource languages is not analyzed in this study thus the need to address the issue of language diversity for real-world applicability. Additionally, the proposed model is not explicitly tested on noisy app reviews that are common in real-world settings. The model lacks preprocessing methods for cleaning noisy app review data, compromising its effectiveness in ABSA tasks due to noisy or misleading information.
3. **Absence of Multimodal Analysis:** The main source of data for analysis in this study was textual data gathered from app reviews. However, RE works traditionally use mixed media, including diagrams and flowcharts, screenshots, audio and video interviews, and user reviews. Diagrams and flowcharts are used to explain systems and workflows, screenshots are used to represent the current state of a system or User Interface (UI), and interviews are used to gather knowledge of the system from stakeholders. Thus, failure to incorporate these additional modalities reduces the practical value of the proposed approach in situations where combining data from various modalities is crucial for a proper comprehension of requirements.
4. **Shortcomings of the Selected Explainability Method:** The use of LIME as an explainability method in this study was based on its simplicity, interpretability, and general applicability to generating local explanations. Thus, it is especially valuable when used in model interpretation because of its feature-level details. However, as indicated before, LIME has some limitations. The output can be inconsistent across multiple runs because the model uses random perturbations that produce varied explanations for the same input. Furthermore, its perturbation-based approach may not uncover the complex and highly dependent structure of current language models, such as BERT. This can result in approximations that are sub-optimal in the true sense related to the actual decision-making of the model, especially with regard to the language understanding mechanism of the model.

Even though the results of our work are encouraging, more investigation is required to overcome the given limitations and look into wider applications. Further, there are opportunities for future research. The following are some aspects worth exploring further.

1. **Scalability Across Diverse Datasets and Benchmark Dataset Expansion:** In future work, we plan to evaluate the proposed

approach using different datasets from various fields, including health care, financial, educational, and e-commerce, to verify the robustness and generalizability of the proposed approach from a requirement engineering perspective. This includes issues of variability in features such as vocabulary, syntactic complexity, and annotation within and across domains. Further, the use of multilingual datasets, especially those focused on low-resource languages, including multimodal content such as diagrams, screenshots, and audio-visual data from interviews and user reviews, will enhance the consideration of practical demands for real-world RE processes. In addition, the AWARE dataset can be further expanded. The AWARE dataset only contains reviews from the Social Networking, Productivity, and Games domains. This can be extended further by including reviews from additional domains. To include app reviews from other domains, various platforms will help gather app reviews, such as app stores, social networking sites, and online forums, where users share their experiences with apps. These app reviews might include the education, finance, travel, entertainment, health, and fitness domains.

2. **Integration of Multi-modal analysis:** A more comprehensive RE framework will be developed in the future by incorporating multimodal sources for requirement elicitation, such as flowcharts, screenshots, diagrams, audio and video interviews, and user reviews, as these sources provide a more holistic view of user experiences. Integrating text, audio, video, and imaging modalities can help to better understand user experience. Instead of just analyzing text-based app reviews, it will be better to perform multimodal analysis to obtain a comprehensive understanding of feedback from end users.
3. **Exploration of advanced XAI methods for better interpretability of model's decisions:** LIME was used in our study because of its simplicity and interpretability. However, LIME suffers from some notable limitations, such as the instability of explanations provided by LIME because it depends on random perturbations, and it might yield approximations that do not perfectly match the internal representations of intricate models such as BERT. Future research will explore advanced explainability methods that offer more stability, accuracy, and compatibility with transformer-based model's internal mechanisms.
    - **SHAP:** The SHAP XAI method (Lundberg, 2017) can be used to achieve consistent and mathematically based feature importance scores, which guarantee stable and trustworthy explanations, especially for token-level contributions in models such as BERT.
    - **Anchors:** Unlike perturbation-based methods, anchors produce interpretable, rule-based local explanations that are more reliable and useful (Ribeiro et al., 2018).
    - **Attention Mechanisms:** By emphasizing the most significant tokens in the decision-making process, the transformer-based model's inherent attention weights offer accurate and transparent explanations and can be utilized for the transformer-based model's interpretability (Vaswani et al., 2017).
4. **User-Centric App Development:** Prioritizing user experience and anticipating user needs expressed in user feedback and reviews are components of user-centric app development. ABSA supports user-centric app development, promoting long-term user satisfaction.
5. **Fine-tuning of LLMS for ABSA tasks:** Fine-tuning of LLMs like BERT (Devlin et al., 2018), RoBERTa (Liu et al., 2019), GPT-3 (Brown et al., 2020), GPT-4 (Achiam et al., 2023), for ABSA tasks enhances their ability to capture contextual meaning hidden in text data by exposing them to domain-specific data that can be user feedback, comments, or review text. This refinement will help prediction models accurately deliver the ABSA results.

6. Conclusion

Our study proposed a transformative approach based on ABSA and XIA. The aim was to accelerate the Requirement Elicitation (RE) process in an automated manner by providing meaningful insights into user preferences and requirements. We significantly improved the RE process by enhancing the performance and robustness of the BERT model using fine-tuning techniques for ABSA of app reviews and introduced XAI using the LIME approach for better interpretation of app review analysis results. Experiments were conducted on the benchmark dataset AWARE, which contains 11,323 app reviews from the Social Networking, Productivity, and Games domains. Our proposed methodology not only outperformed existing baseline studies but also offered meaningful insights into users' preferences and attitudes towards mobile apps. Specifically, our fine-tuned BERT model exhibited outstanding performance on ACD and ACP tasks across all the AWARE dataset domains. For the ACD task, we attained the highest F1-Score of 0.78 for the AWARE Social Networking domain, 0.82 for the Aware Productivity domain, and 0.89 for the Aware Games domain. For the ACP task, our model also achieved an impressive accuracy score of 0.92 for the AWARE Social Networking domain, 0.92 for the AWARE Productivity domain, and 0.96 for the AWARE Games domain. These results highlight the robustness and effectiveness of the proposed method.

Additionally, our proposed methodology employed LIME, an XAI technique, to obtain post-hoc explanations for the ABSA results obtained from our fine-tuned BERT classification model. These post hoc explanations were in the form of highlighted text segments, which were deemed most important for the classification of reviews for the ACD and ACP tasks. The prediction probabilities of the possible potential classes were also visualized using LIME. LIME derived these prediction probabilities by calculating the level of influence of each highlighted word on the classifier's prediction. LIME also provides local explanations showing how influential text elements impact the ABSA process overall, allowing a deeper understanding of the models decision-making process. We also examined the robustness of our fine-tuned BERT model and the stability and reliability of the LIME model predictions by calculating the Local Fidelity and Perturbation Stability scores of five random samples from the AWARE dataset for each ABSA task. We achieved higher values of Local Fidelity and Perturbation stability scores that clearly signify the robustness of our fine-tuned BERT model and the credibility of the LIME explanations. Thus, from the above results and observations, we conclude that our proposed methodology based on the ABSA of app reviews using BERT fine-tuning and then integration of XAI using the LIME approach for post hoc explanations collectively aids the RE process by automating the requirement elicitation process using app review analysis to obtain better insights into user preferences. Ultimately, this contributes to the development of user-centric applications.

CRediT authorship contribution statement

Soonh Taj: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Sher Muhammad Daudpota:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization. **Ali Shariq Imran:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Zenun Kastrati:** Writing – review & editing, Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the Department of Computer Science (IDI), Faculty of Information Technology and Electrical Engineering, Norwegian University of Science and Technology (NTNU), Gjøvik, Norway; and in part by the Curricula Development and Capacity Building in Applied Computer Science for Pakistani Higher Education Institutions (CONNECT) Project NORPART-2021/10502, funded by Norwegian Directorate for Higher Education and Skills (DIKU).

Data availability

Data will be made available on request.

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